CSA SAR Imaging: Simulation Method

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Contents

1 Overview

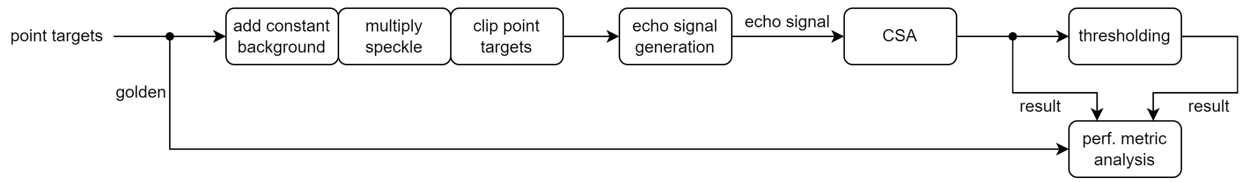


Figure 1: Simulation flow overview.

2 Simulation Setup

2.1 Signal Model

2.1.1 Chirp Echo Signal

The multi-target scattering model of the chirp echo signal is represented as a matrix $S = [s_{m,n}]_{M\times N}$, where:

- $\eta = m\Delta\eta$: slow time
- $\tau = n\Delta\tau$: fast time
- A_k: scattering coefficient of point target k
- $R_k(m\Delta\eta) = \sqrt{(R_0 + \Delta R_{0,k})^2 + (m\Delta\eta + \Delta\eta_k)^2 V_r^2}$, where
  - $\Delta R_{0,k}$: slant-range offset from target k to scene centre
  - $\Delta\eta_k$: azimuth time offset from target k to scene centre

$$s_{m,n} = \sum_k A_k \operatorname{rect}\left(\frac{n\Delta\tau - \frac{2R_k(m\Delta\eta)}{c}}{T_r}\right) \cdot e^{j\pi\frac{B_r}{T_r}\left(n\Delta\tau - \frac{2R_k(m\Delta\eta)}{c}\right)^2} \cdot e^{-j4\pi\frac{R_k(m\Delta\eta)}{\lambda}} \tag{1}$$

2.1.2 Coherent Scattering

Time-Domain Model Coherent scattering applied in the time domain:

$$S \circ C + W, \quad C = [c_{m,n}]_{M\times N}, \quad c_{m,n} = e^{j\theta_{m,n}}, \quad \theta_{m,n} \sim \mathcal{U}(0, 2\pi) \tag{2}$$

Spatial-Domain Model Coherent scattering applied in the spatial (image) domain:

$$T(S + W) \circ C, \quad c_{m,n} \sim \mathcal{E}(1) \tag{3}$$

where $T(\cdot)$ denotes the imaging algorithm (e.g. RDA or CSA).

2.1.3 Thermal Noise

Thermal noise is modelled as a complex Gaussian matrix $W = [w_{m,n}]_{M \times N}$ with $w_{m,n} \sim \mathcal{CN}(0, \sigma)$:

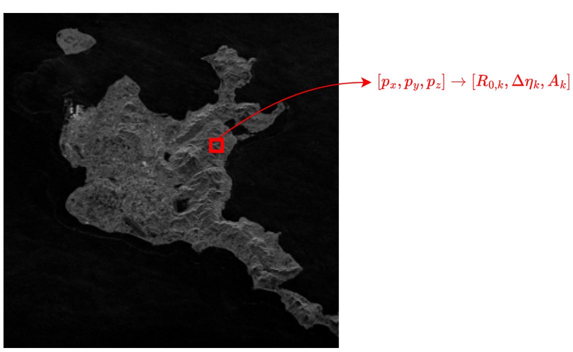
$$\begin{aligned}
\sigma^2 &= P_n/2 && (4)\\
P_n &= P_s/\mathrm{SNR} \quad \text{(instantaneous noise power)} && (5)\\
P_s &= \Re\{x[n]\}^2 + \Im\{x[n]\}^2 \quad \text{(instantaneous signal power)} && (6)
\end{aligned}$$

2.2 Image-to-Echo Conversion

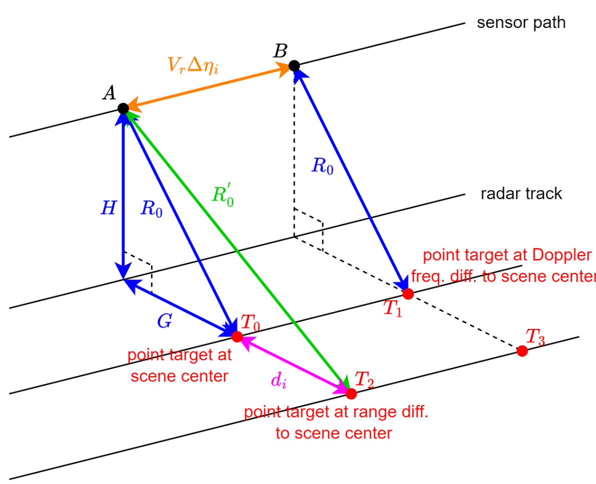
Given a pixel displacement d_i in the ground-range direction and the geometry $R_0^2 = H^2 + G^2$, the slant-range to the displaced pixel is $R_0' = \sqrt{H^2 + (G + d_i)^2}$. A Taylor expansion gives:

$$\begin{aligned}
R_0' - R_0 &= \sqrt{H^2 + (G + d_i)^2} - R_0 \\
&= \sqrt{R_0^2 + 2Gd_i + d_i^2} - R_0 \\
&\approx R_0\left(1 + \frac{2Gd_i + d_i^2}{2R_0^2}\right) - R_0 \quad \text{(first-order Taylor)} \\
&= \frac{2Gd_i + d_i^2}{2R_0} \\
&\approx \frac{Gd_i}{R_0} \quad (d_i \ll G) \\
&\approx d_i \quad (G \approx R_0) && (7)
\end{aligned}$$

Therefore $R_0' \approx R_0 + d_i$.



(a) Image-to-echo conversion.



(b) Multi-point target model.

2.3 Simulation Parameters

Table 1: Simulation parameter definitions.

Parameter	Description
`wavelength_m`	Carrier wavelength (m)
`pulse_width_sec`	Pulse duration (s)
`pulse_rep_freq_hz`	Pulse repetition frequency (Hz)
`bandwidth_hz`	Range chirp bandwidth (Hz)
`sampling_freq_hz`	Range sampling frequency (Hz)
`closest_slant_range_m`	Closest slant range to scene centre (m)
`azi_win_en`	Enable azimuth window (beam pattern)
`rng_pad_time`	Range padding factor; row length = $T_r \cdot f_s \cdot$`rng_pad_time`
`noise_en`	Enable additive thermal noise
`snr_db`	SNR in dB (ignored when `noise_en` is `False`)

3 Performance Metrics

All metrics are computed after converting both images to dB scale and applying peak-to-peak−30 dB normalisation to $[0, 1]$.

3.1 Structural Similarity Index (SSIM)

$$\text{SSIM} = L \cdot C \cdot S \tag{8}$$

Luminance

$$L(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \tag{9}$$

Compares local mean brightness; $C_1 = (0.01 \cdot d)^2$.

Contrast

$$C(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \tag{10}$$

Compares local standard deviation; $C_2 = (0.03 \cdot d)^2$.

Structure

$$S(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \tag{11}$$

Normalised cross-covariance (Pearson correlation of patch values); $C_3 = C_2/2$.

3.2 PSNR and MSE

$$\mathrm{PSNR} = 20\log_{10}\left(\frac{1}{\sqrt{\mathrm{MSE}}}\right)\ \mathrm{dB}, \qquad \mathrm{MSE} = \frac{1}{N}\sum_i (x_i - y_i)^2 \tag{12}$$

3.3 Signal-to-Clutter Ratio (SCR)

$$\mathrm{SCR} = 10\log_{10}\left(\frac{\bar{P}_{\text{signal}}}{\bar{P}_{\text{clutter}}}\right)\ \mathrm{dB} \tag{13}$$

The brightest pixel is taken as the target centre. An inner disk of radius $r_1 = 5\,\mathrm{px}$ defines the *signal zone*; an annulus between r_1 and $r_2 = 15\,\mathrm{px}$ defines the *clutter zone*.

3.4 Image Entropy

$$E = -\sum_{i=1}^{M}\sum_{j=1}^{N} P_{i,j}\ln P_{i,j}, \qquad P_{i,j} = \frac{|I_{i,j}|^2}{\sum_{i,j}|I_{i,j}|^2} \tag{14}$$

where $\lim_{P\to 0} P\ln P = 0$ by convention.

4 Results

4.1 Metric Summary

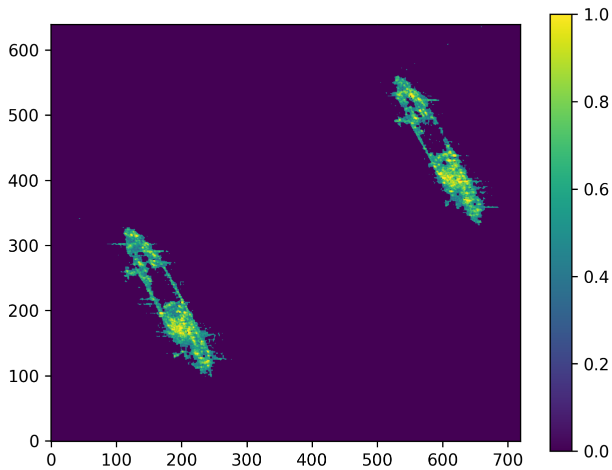
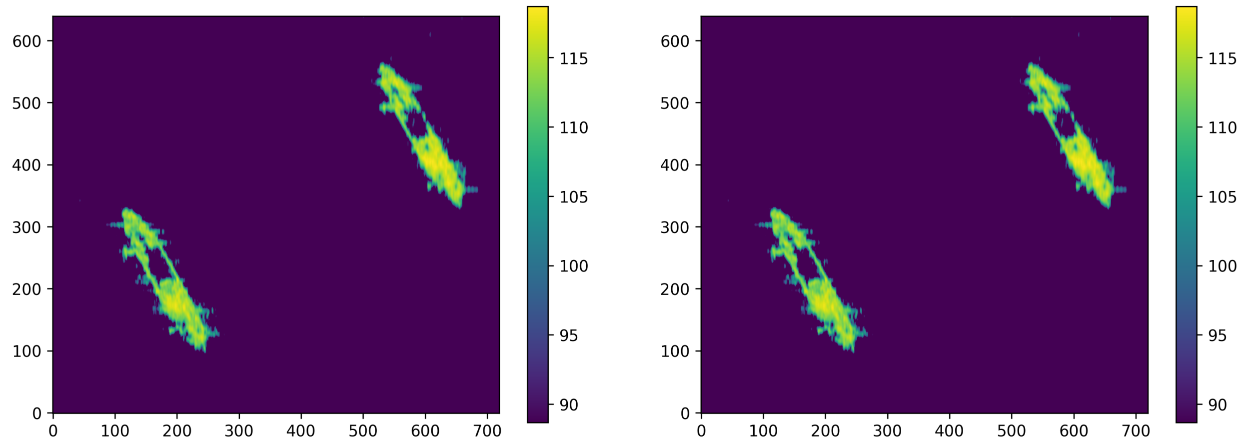


Figure 3: Golden image.



(a) Result (threshold = 0).

(b) Result (threshold = 10^4).

Figure 4: CSA imaging results.

Table 2: Performance metrics.

Metric	result (threshold = 0)	result (threshold = 1e4)
SSIM	0.09735	0.09737
Luminance L	0.29189	0.29023
Contrast C	0.13559	0.13411
Structure S	0.94457	0.95153
PSNR (dB)	14.44	14.43
MSE $[0, 1]$	0.03600	0.03605

4.2 SSIM Component Interpretation

Table 3: SAR interpretation of each SSIM component.

Component	SAR Interpretation	Role in Object Detection
Luminance L	Average backscatter	Distinguishes overall brightness; secondary to shape.
Contrast C	Variance / intensity	Separates target signal from background clutter.
Structure S	Spatial correlation	Primary metric; defines shape, prevents false alarms.

A Umbra Image Denoising

A.1 Processing Flow

All methods start by normalising the float SAR slice to uint8 $[0, 255]$ via min-max scaling, which is required by the OpenCV processing functions. The pipeline then diverges by method:

- **Raw** — no processing; the normalised slice is used as-is.
- **Binary** — a fixed threshold is applied; pixels above the threshold are set to 1, others to 0.
- **Otsu** — same as binary, but the threshold is computed automatically by minimising intra-class intensity variance.
- **Filter methods** (Bilateral, Guided, TV, Lee) — a denoising filter is applied first to suppress speckle, then Otsu thresholding is run on the smoothed image to produce a cleaner binary mask.

For filter methods, the role of the filter is to improve the quality of the subsequent Otsu threshold: speckle in raw SAR creates a noisy intensity histogram, causing Otsu to produce a ragged mask. Denoising first yields two cleaner histogram peaks (target vs. clutter) so the threshold is more reliable.

After thresholding, morphological closing is optionally applied to fill small holes inside detected regions.

A.2 Morphological Closing

Morphological closing is a two-step operation applied to a binary image:

1. **Dilation** — expands bright regions by adding pixels around their edges.
2. **Erosion** — shrinks the dilated regions back by removing pixels around their edges.

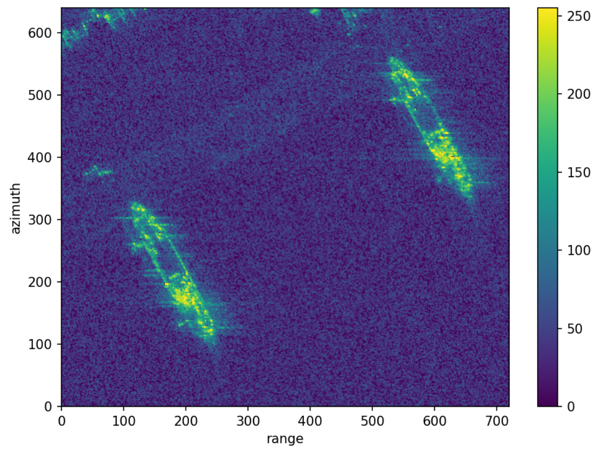
The net effect is that small gaps and holes within detected regions are filled and nearby disconnected blobs are merged, while the overall shape and size remain roughly unchanged. Formally, closing of image I by structuring element B is defined as:

$$I \bullet B = (I \oplus B) \ominus B \tag{15}$$

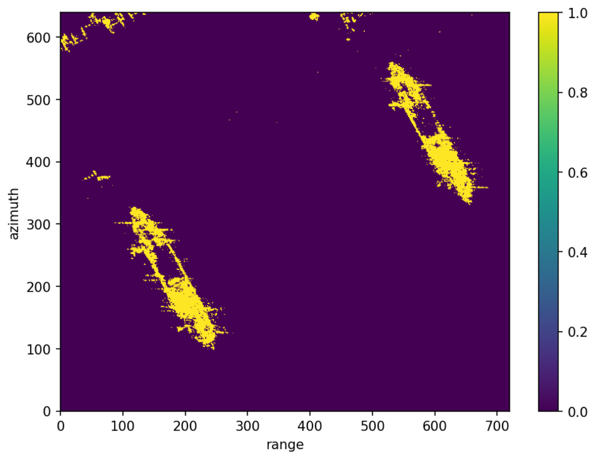
where $\oplus$ denotes dilation and $\ominus$ denotes erosion.

In the SAR context, after thresholding or denoising produces a rough binary mask, morphological closing fills small dark holes inside a detected target (e.g. a ship) that were incorrectly zeroed out, yielding a more solid and contiguous detection region.

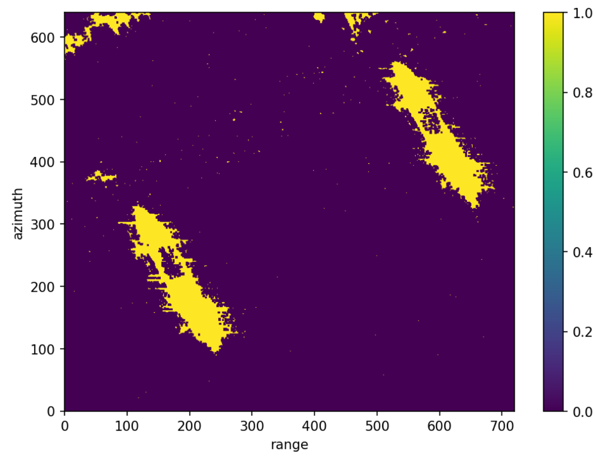
A.3 Results



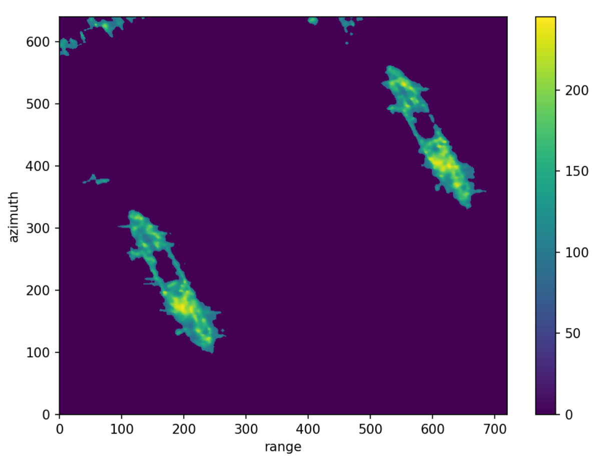
(a) Raw image.



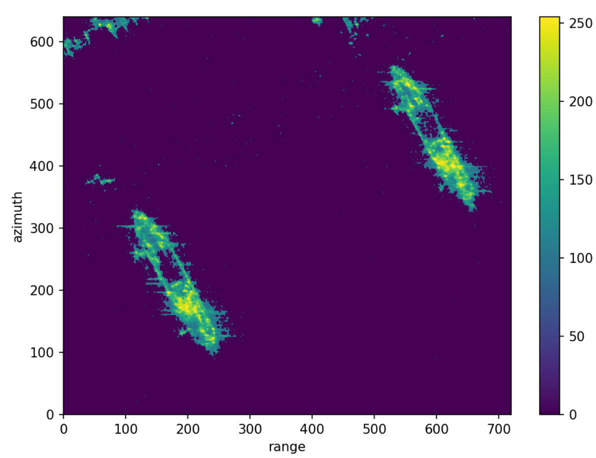
(a) Binary thresholding.



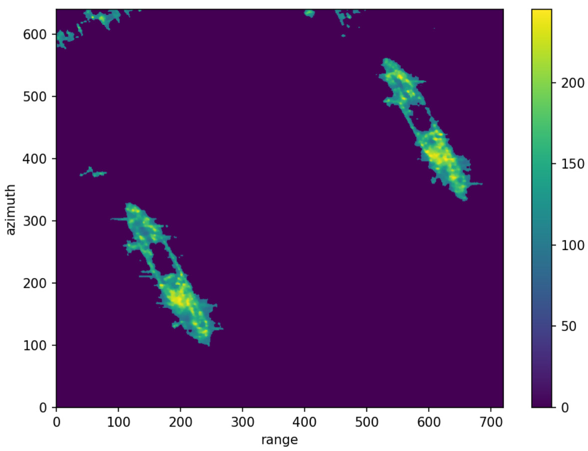
(b) Otsu thresholding.



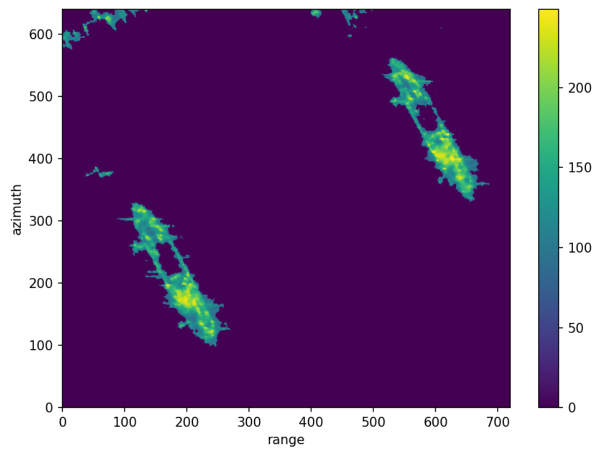
(a) Bilateral filter.



(b) Guided filter.



(a) Total variation filter.



(b) Lee filter.

Table 4: Non-zero sample counts per denoising method.

Method	Non-zero samples
Raw SAR Image	443,448
Binary Thresholding (threshold = 110)	16,419
Otsu's Thresholding + Morphological Closing	27,572
Bilateral Filter + Morphological Closing	23,629
Guided Filter + Morphological Closing	27,087
Total Variation (TV) Filter + Morphological Closing	23,075
Lee Filter + Morphological Closing	23,873